# Longitudinal Progression of Motor Symptoms in Parkinson's Disease - Group 11

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**Declaration: We declare that all students in the group have contributed sufficiently to the study and that the work is original and non copied from elsewhere.**

## Introduction

The aim of this analysis is to understand the evolution of motor symptom severity over time in patients diagnosed with Parkinson's Disease (PD), with a primary focus on the role of smoking status. Motor symptom severity is assessed using the Unified Parkinson's Disease Rating Scale (UPDRS) Part III, which ranges from 0 to 108, where higher scores indicate more severe motor impairment. In addition to smoking, demographics such as age at diagnosis and sex are considered as potential influencing factors. The study includes 500 patients diagnosed between 1995 and 2011, followed longitudinally with assessments at baseline and annually for seven consecutive years. For each patient, UPDRS scores, visit timing, smoking history at baseline, age, and sex were recorded.

The main research question guiding this analysis is: How does motor symptom severity (UPDRS Part III) evolve over time in Parkinson's patients, and to what extent is this evolution influenced by smoking status, after accounting for age and sex?

We will start the report with a Descriptive Analysis section, examining the variable distributions and bivariate relationships, exploring the mean, variance, and correlation structure to detect main tendencies and eventual non-linearities. The information gathered will inform the subsequent Statistical Analysis, in which we will address the research question by building a linear mixed-model and presenting out conclusions. Finally, in the Appendix we will include more extensive details and visualizations to support our case.

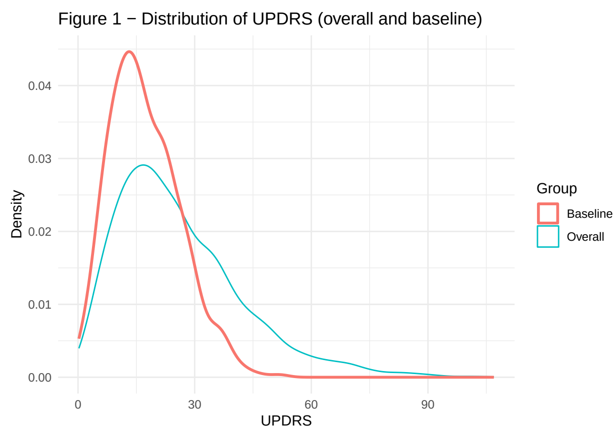
## 1 Descriptive Statistics

### 1.1 Summary Statistics for the Sample

The study population consisted of 500 participants with a wide age range (41.1-88 years), although the median age was above 65 years, indicating an overall older cohort. The distribution of sex was balanced, with equal numbers of male and female participants. Smoking status was unevenly distributed across groups, with more than half of the participants classified as never smokers (52.2%), followed by former smokers (28.2%) and a smaller proportion of current smokers (19.6%). The relatively small size of the current smoker group may limit precision when comparing disease trajectories across smoking categories, which will be considered in the subsequent analysis.

Table 1: Descriptive Statistics

| Variable | Value |
|---|---|
| Age (years), mean (SD) | 64.64 (13.49) |
| Age (years), median | 65.1 |
| Age (years), min–max | 41.1 – 88 |
| Sex: Male, n (%) | 250 (50%) |
| Sex: Female, n (%) | 250 (50%) |
| Smoking status: Never (0), n (%) | 261 (52.2%) |
| Smoking status: Former (1), n (%) | 141 (28.2%) |
| Smoking status: Current (2), n (%) | 98 (19.6%) |




Across all visits, UPDRS scores (see Figure 1) exhibited a right-skewed distribution, with most observations concentrated in the mild-to-moderate range and a long tail toward higher values. The median score was 22.73, with values ranging from 0.22 to 106.94, indicating a wide range of motor severity across the study period. At baseline, UPDRS scores were lower and less variable, with a median of 16.02 and a narrower range (0.23–51.85), suggesting that most patients entered the study with mild to moderate impairment. The baseline distribution also showed moderate variability at diagnosis and a right skew. This skewness is noted as it may influence model fit and will be evaluated through residual diagnostics in the Statistical Analysis section.

The age distribution was broadly similar across sex groups, with comparable medians (65.2 years for males and 64.1 years for females) and overall variability. No substantial age differences in variability were observed across sex groups(see Appendix A). Similarly, age distributions were comparable across smoking groups, with substantial overlap between categories. Former smokers appeared slightly older on average, while current smokers were marginally younger; however, these differences were small (see Appendix B).

## 1.2 Progression of Disease Severity (UPDRS)

The relationship between age and UPDRS (see Appendix A2) showed a weak positive trend, with substantial variability across all ages. Visual inspection suggests mild non-linearity, with increasing spread in UPDRS values at higher ages and a notable presence of outliers. In contrast, mean UPDRS trajectories over time were highly similar across sex groups (see Appendix A3), with no clear separation between males and females. However, heterogeneous progression patterns across individuals indicate variability that may affect assumptions of linearity and homoscedasticity. Together, these findings do not indicate strong confounding by sex, while age may act as a potential confounder in the relationship between smoking status and disease progression.

Figure 2.1 − Individual UPDRS trajectories with mean trend

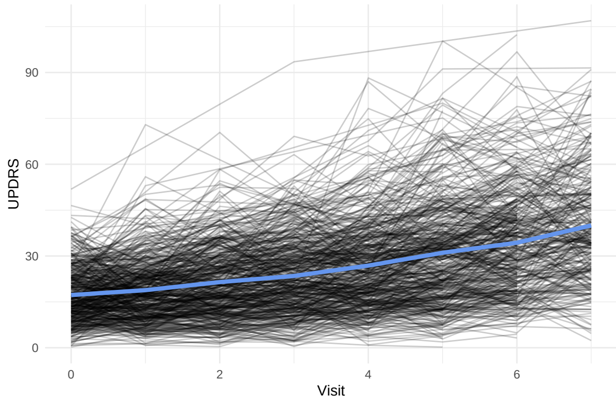



Figure 2.2 − Mean UPDRS trajectory with 95% CI

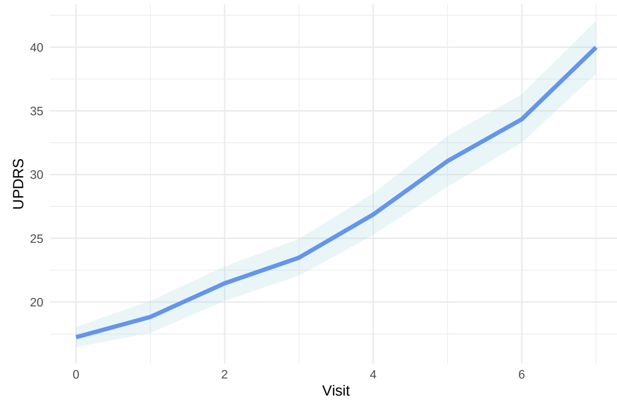



Figure 3 − Mean UPDRS over time by smoking status

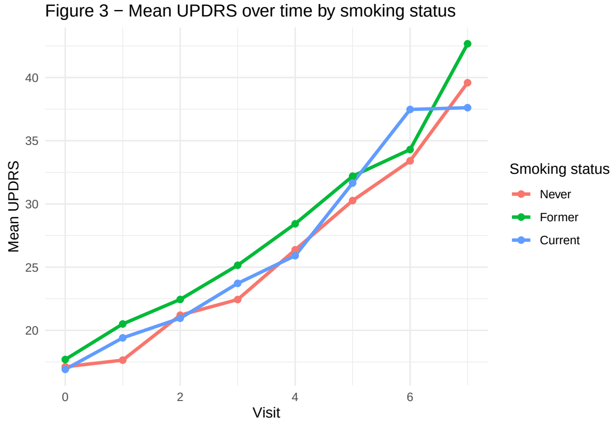
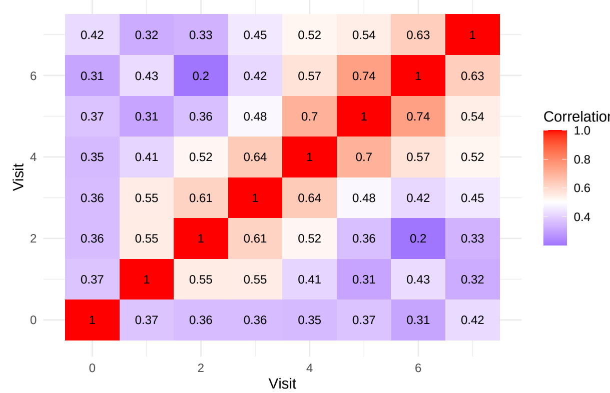



Figure 4 − Correlation of UPDRS across visits




The longitudinal distribution of UPDRS scores indicates a clear increase in mean values over time, as illustrated in Figure 2.1, reflecting progressive motor impairment. While the overall trend appears approximately linear, systematic variations in the rate of increase across visits suggest a mild non-linear pattern consistent with a potential quadratic time effect. Substantial variability is observed both within and between patients, as illustrated by the divergent individual trajectories around the overall mean trend. Confidence intervals around the mean trajectory (Figure 2.2) widen over time, indicating increasing variability that may violate the assumption of homoscedasticity. This indicates heterogeneity in disease progression across individuals, as well as within-subject fluctuations over time.

The correlation matrix of UPDRS measurements across visits indicates moderate to strong positive correlations between repeated observations within individuals, confirming the presence of within-subject dependence. Correlations tend to be higher for visits closer in time and lower for more distant visits, although the pattern is not strictly monotonic. This suggests a temporally structured correlation, where measurements within the same individual are related over time. These findings motivate the use of mixed-effects models, which account for such correlation through subject-specific random effects, including both random intercepts and slopes.

## Statistical Analysis

```
model_final <- lme(
  UPDRS ~ Visit + I(Visit^2) + smoking_status + Visit:smoking_status + Age + sex,
  random = ~ Visit | id,
  data = DF,
  method = "ML",
  na.action = na.exclude,
```

```
  weights = varExp(form = ~ Visit)
)
```

#### Assumptions

## Appendix

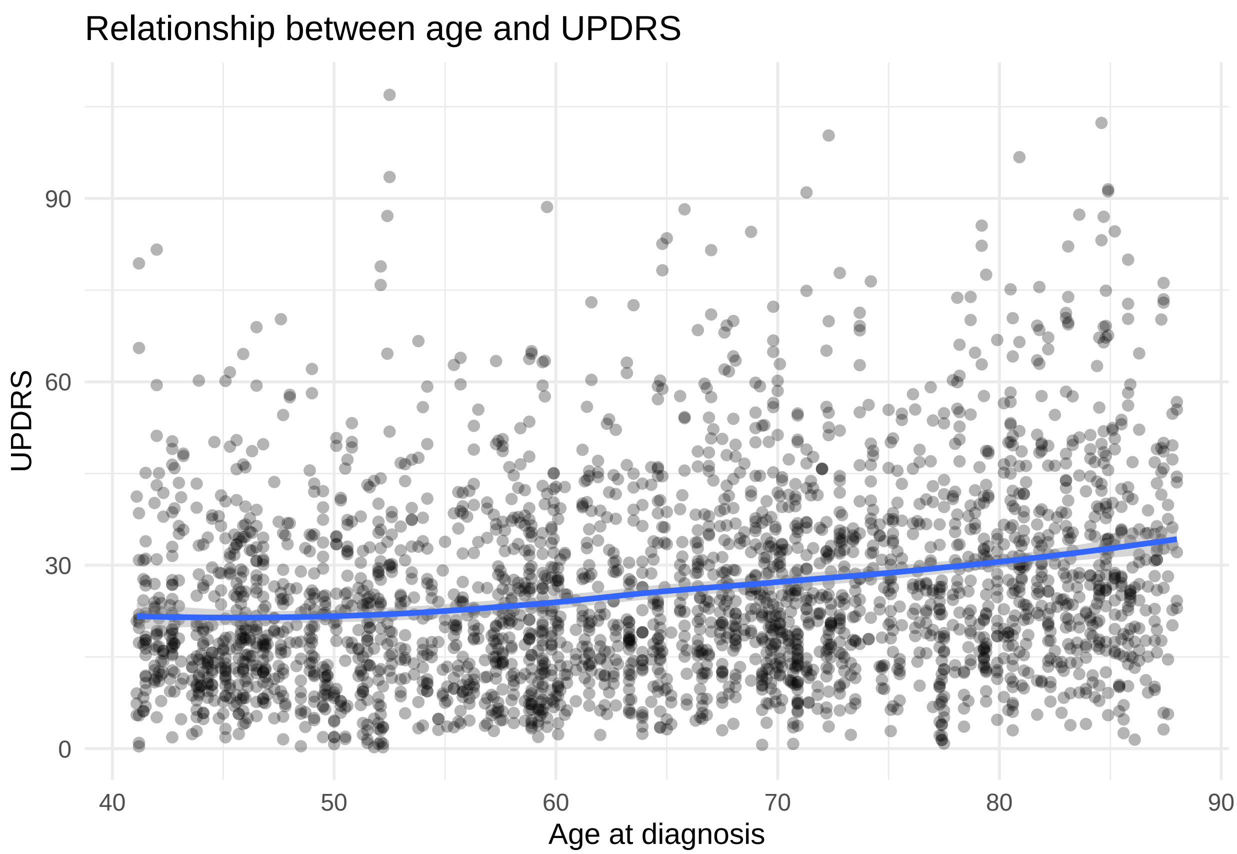
**Appendix A1: Relationship between Age and Sex** This plot describes the distribution of Age by Sex.

```
ggplot(DF_pat, aes(x = sex, y = Age, fill = sex)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.3) +
  labs(
    title = "Age distribution by sex",
    x = "Sex",
    y = "Age"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

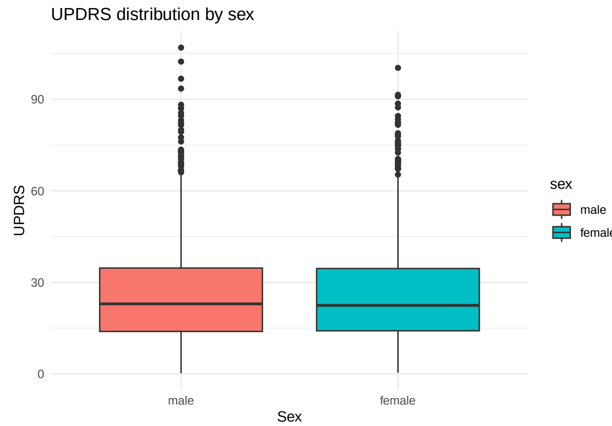

**Appendix A2: Relationship between UPDSR and Age**

```
ggplot(DF, aes(x = Age, y = UPDRS)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "loess") +
  labs(
    title = "Relationship between age and UPDRS",
    x = "Age at diagnosis",
    y = "UPDRS"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

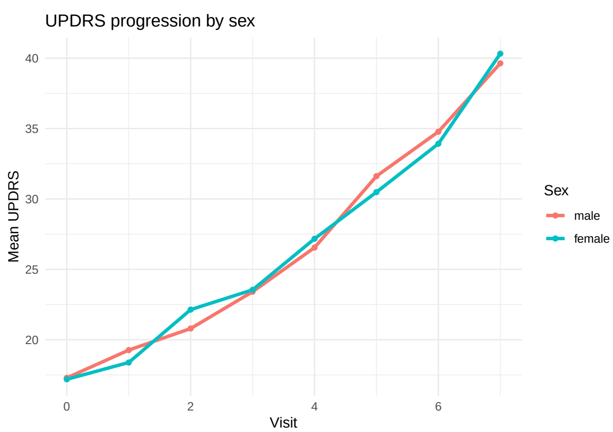



```
ggplot(DF,
       aes(x = sex, y = UPDRS, fill = sex)) +
  geom_boxplot() +
  labs(
    title = "UPDRS distribution by sex",
    x = "Sex",
    y = "UPDRS"
  ) +
  theme_minimal()
```




### Appendix A3: Relationship between UPDRS and Sex

```
ggplot(DF,
       aes(x = Visit, y = UPDRS, color = sex)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1.2) +
  stat_summary(fun = mean, geom = "point") +
  labs(
    title = "UPDRS progression by sex",
    x = "Visit",
    y = "Mean UPDRS",
    color = "Sex"
  ) +
  theme_minimal()
```

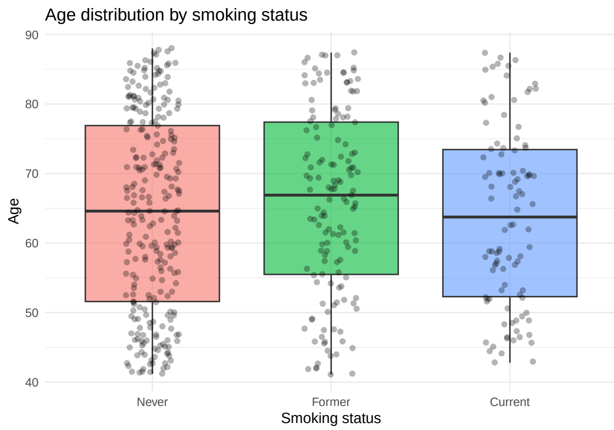



**Appendix A2: Relationship between Age and Smoking Status** This plot describes the distribution of Age by Smoking Status.

```
DF_pat = DF %>%
  distinct(id, .keep_all = TRUE)

ggplot(DF_pat, aes(x = smoking_status, y = Age, fill = smoking_status)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.3) +
  labs(
    title = "Age distribution by smoking status",
    x = "Smoking status",
```

```r
    y = "Age"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```




**Appendix B: Model specification, evaluation, and selection** We applied a systematic model building strategy, proceededing in a stepwise manner. We started from a base mixed-effects model with a linear time effect and constant residual variance. Given indications from the descriptive analysis of increasing variability over time, a variance function (`varExp`) was introduced to allow for heteroscedasticity. In parallel, potential non-linear effects of time were evaluated using both natural splines and a quadratic specification. Models were compared using likelihood ratio tests, AIC, and BIC. The inclusion of a variance structure led to a substantial improvement in model fit, while both spline-based and quadratic time effects improved fit relative to the linear specification. However, no meaningful difference was observed between spline and quadratic formulations, and the quadratic model was preferred based on parsimony. The final model therefore included a quadratic time effect, a time-dependent variance structure, and subject-specific random intercepts and slopes.

```r
library(nlme)
library(splines)

# Model 1: Base model (linear time, constant variance)
model1 <- lme(
  UPDRS ~ Visit * smoking_status + Age + sex,
  random = ~ Visit | id,
  data = DF,
  method = "ML",
  na.action = na.exclude
)

# Model 2: Add variance structure (heteroscedasticity over time)
model2 <- lme(
  UPDRS ~ Visit * smoking_status + Age + sex,
  random = ~ Visit | id,
  data = DF,
  method = "ML",
  na.action = na.exclude,
  weights = varExp(form = ~ Visit)
)
```

```r
# Model 3: Add non-linearity using natural spline
model3 <- lme(
  UPDRS ~ ns(Visit, df = 3) * smoking_status + Age + sex,
  random = ~ Visit | id,
  data = DF,
  method = "ML",
  na.action = na.exclude
)

# Model 4: Spline + variance structure
model4 <- lme(
  UPDRS ~ ns(Visit, df = 3) * smoking_status + Age + sex,
  random = ~ Visit | id,
  data = DF,
  method = "ML",
  na.action = na.exclude,
  weights = varExp(form = ~ Visit)
)

# Model 5: Quadratic time effect
model5 <- lme(
  UPDRS ~ Visit + I(Visit^2) + smoking_status + Visit:smoking_status + Age + sex,
  random = ~ Visit | id,
  data = DF,
  method = "ML",
  na.action = na.exclude
)

# Model 6: Quadratic time + variance structure (final candidate)
model6 <- lme(
  UPDRS ~ Visit + I(Visit^2) + smoking_status + Visit:smoking_status + Age + sex,
  random = ~ Visit | id,
  data = DF,
  method = "ML",
  na.action = na.exclude,
  weights = varExp(form = ~ Visit)
)

# Model comparisons
AIC(model1, model2, model3, model4, model5, model6)
```

```
##        df      AIC
## model1 12 21445.76
## model2 13 21340.16
## model3 18 21422.96
## model4 19 21318.28
## model5 13 21419.15
## model6 14 21311.94
```

```r
BIC(model1, model2, model3, model4, model5, model6)
```

```
##        df      BIC
```

```
## model1 12 21516.85
## model2 13 21417.17
## model3 18 21529.59
## model4 19 21430.83
## model5 13 21496.16
## model6 14 21394.87
```

```r
anova(model1, model2)
```

```
##        Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## model1     1 12 21445.76 21516.85 -10710.88
## model2     2 13 21340.16 21417.17 -10657.08 1 vs 2 107.6004  <.0001
```

```r
anova(model1, model3)
```

```
##        Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## model1     1 12 21445.76 21516.85 -10710.88
## model3     2 18 21422.96 21529.58 -10693.48 1 vs 2 34.80504  <.0001
```

```r
anova(model1, model5)
```

```
##        Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## model1     1 12 21445.76 21516.85 -10710.88
## model5     2 13 21419.15 21496.16 -10696.58 1 vs 2 28.60864  <.0001
```

```r
anova(model2, model4)
```

```
##        Model df      AIC      BIC    logLik   Test L.Ratio p-value
## model2     1 13 21340.16 21417.17 -10657.08
## model4     2 19 21318.28 21430.83 -10640.14 1 vs 2 33.8825  <.0001
```

```r
anova(model2, model6)
```

```
##        Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## model2     1 13 21340.16 21417.17 -10657.08
## model6     2 14 21311.94 21394.87 -10641.97 1 vs 2 30.22059  <.0001
```

```r
anova(model4, model6)
```

```
##        Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## model4     1 19 21318.28 21430.83 -10640.14
## model6     2 14 21311.94 21394.87 -10641.97 1 vs 2 3.661907   0.599
```

**Appendix C: Model Diagnostics**